

Aditya Kaki

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EDUCATION

Virginia Polytechnic Institute and State University (Virginia Tech)

Blacksburg, VA

B.S. Computer Engineering, Machine Learning Concentration

Expected May 2027

- Relevant Coursework: Embedded Systems, Computer Architecture, Machine Learning, Artificial Intelligence, Signals and Systems, Applied Software Design

EXPERIENCE

Founder & Technical Lead

Mar 2026 – Present

ARK Integrity Systems

Blacksburg, VA

- Building ARK, a secure video integrity platform that cryptographically signs CCTV footage at the moment of capture using EAL4+ Common Criteria certified TPM 2.0 hardware key storage and ECDSA P-256, creating forensic records that hold up to courtroom scrutiny
- Designing an embedded bridge device that retrofits existing IP camera systems with on premise cryptographic signing at the edge, removing the risk of post capture tampering without needing to replace deployed hardware
- Developing embedded Linux firmware for an Nvidia Jetson Orin Nano bridge, handling RTSP stream interception, Merkle tree frame hashing, and TPM backed signing across multiple camera streams, while bringing up a hardware NVDEC decode path and CUDA/TensorRT accelerated YOLOv8 inference
- Built a open source verification tool that re-derives Merkle roots and validates ECDSA signatures and RFC 3161 trusted timestamps, so any attorney, court, or insurance adjuster can independently confirm footage authenticity
- Running customer discovery conversations with small business owners, civil attorneys, and insurance adjusters to validate product market fit and shape pricing strategy

Digitalization Intern

May 2025 – Aug 2025

TITAN America – Corporate Engineering

Troutville, VA

- Built Python automation that generated a nested engineering asset directory structure of over 5,000 folders from Excel source data, standardizing infrastructure for enterprise digital records
- Developed an OCR powered document classification pipeline with Tesseract and Python to process, orient, and route hundreds of technical engineering documents into the correct asset locations
- Designed and populated structured digital infrastructure that corporate engineering and maintenance planning teams now use as a foundation for internal AI workflow initiatives, including the AI Planner and AI Librarian projects

PROJECTS

Machine Learning Pathogen Classification | *PyTorch, XGBoost, scikit-learn, pandas*

Jan 2026 – May 2026

- Built a team based machine learning pipeline for multi class pathogen classification using 13 public GEO transcriptomic datasets covering more than 1,700 clinical patient samples
- Developed preprocessing, GEO data ingestion, iPAGE feature engineering, LASSO based feature selection, and leave one dataset out validation workflows to support robust cross cohort model evaluation
- Trained and evaluated PyTorch neural networks, XGBoost, Random Forest, and Logistic Regression models to classify bacterial, viral, and healthy patient samples
- Reached a macro F1 score above 0.53 and an AUROC above 0.94 in the strongest cohorts

Mechanical Energy Harvester | *Piezoelectric Transducers, Circuit Design*

Jan 2025 – Feb 2025

- Designed a vibration powered energy harvesting prototype using piezoelectric transducers and rectification circuitry to convert mechanical motion into stored DC energy
- Tested energy capture performance across human motion and mechanical vibration scenarios through PZT based transducer optimization

TECHNICAL SKILLS

Programming: Python, C, C++, Java, Verilog

ML/Data: PyTorch, scikit-learn, XGBoost, pandas, NumPy, Tesseract OCR

Systems/Tools: Linux, Git, MATLAB, Quartus, Arduino IDE, VS Code

Hardware: Raspberry Pi CM4, TPM 2.0, Arduino, PCB Prototyping, Breadboard Circuit Design, DE-10 Lite FPGA Board, NFC/RFID, Embedded System Prototyping